Lab Thirteen – Statistical Inference

1 Remaining Portfolio Work!

By 5p 4/27 Email me a link to your GitHub page with all polished labs rendered, and your review material (at least one unit). In the email, you should include a reflection about the processes involved in creating this portfolio and what you learned as a result. If you're having trouble reflecting, consider Fink's Taxonomy of Significant Learning (Google Fink taxonomy aligned self-reflection questions). In the email, let me know whether you would be comfortable sharing your page – I'm hoping that together we can get a good study guide for the final.

2 Statistical Inference

Kasdin et al. (2025) show that dopamine in the brains of young zebra finches acts as a learning signal, increasing when they sing closer to their adult song and decreasing when they sing further away, effectively guiding their vocal development through trial-and-error. This suggests that complex natural behaviors, like learning to sing, are shaped by dopamine-driven reinforcement learning, similar to how artificial intelligence learns. You can find the paper at this link: https://www.nature.com/articles/s41586-025-08729-1.

Note that the measurement is rather complicated. First, the measurements are taken on syllables of a bird song. When discussing bird songs, the term syllable refers to a single, distinct, uninterrupted sound that is part of a possibly complex bird song.

They initially describe their measurement as follows. They compare the sound of the developing zebra finches to the median sound of adult zebra finches, that act as a "goal" sound.

> "For each syllable rendition, we calculated the proximity (Euclidean distance) to the median of the adult renditions of the syllable (age > 90 days) in feature space (Fig. 2a,b)."

They contextualize that measurement by computing a "relative quality" that takes into account their current ability (e.g., are they getting closer or further compared to their previous 14 renditions).

> "Because dopamine is hypothesized to encode the relative value of performance, we defined the 'relative quality' of a syllable as proximity to the adult version of the syllable relative to the mean of the previous 14 renditions (Methods)."

They define the "closer" measurements as the 10% of syllable renditions with the highest relative quality and the "further" as the lowest 10%.

> "To test for online error responses, for each day we compared the dopamine activity associated with the
> 10

Note that the data are measures of dopamine change, they use fibre photometry, changes in the fluorescence indicate dopamine changes in realtime. Their specific measurement considers the percent change in flourescence in 100-ms windows between 200 and 300 ms from the start of singing, averaged across development.

Finally, the researchers note that they adjust the observations to correct for the increased dopamine for singing at all (whether it is close or far).

> "As previously reported, the baseline level of dopamine in Area X is known to increase during singing; to isolate the effect of syllable quality from these singing-related dopamine increases, we subtracted the mean dopamine response across all renditions of a syllable on each day from the dopamine response for each rendition."

So, the random variables we are working with are:

$$X_{\text{close}} = \text{average \% change in fluorescence for the top 10\% syllable renditions}$$
$$\text{relative to distance from adult sound (corrected for singing)}$$
$$X_{\text{far}} = \text{average \% change in fluorescence for the bottom 10\% syllable renditions}$$
$$\text{relative to distance from adult sound (corrected for singing)}$$

That is, the unit of measure is the syllable as measured on six birds, and the measurement is somewhat complicated.

2.1 Part a

Using the `pwr` package for `R` (Champely 2020), conduct a power analysis. How many observations would the researchers need to detect a moderate-to-large effect ($d = 0.65$) when using $\alpha = 0.05$ and default power (0.80) for a two-sided one sample t test.

Solution

The researchers need at least 21 observations.

```
library(pwr)
pwr.t.test(d = 0.65,
           power = 0.80, alternative = "two.sided", type = "one.sample")
```

```

     One-sample t test power calculation

              n = 20.58039
              d = 0.65
      sig.level = 0.05
          power = 0.8
    alternative = two.sided
```

2.2 Part b

Click the link to go to the paper. Find the source data for Figure 2. Download the Excel file. Describe what you needed to do to collect the data for Figure 2(g). Note that you only need the `closer_vals` and `further_vals`. Ensure to `mutate()` the data to get a difference (e.g., `closer_vals - further_vals`).

Solution

I downloaded the source data and exported it to CSV. Mac allows me to create a file for each table. I then combined the two columns of the `Closer_vals` and `Farther_vals` into 1 CSV file, named each columnn, and moved the file into Lab13 folder. I constructed a tibble using this CSV file and added a column that captures the difference.

```
dat.birds = read.csv("Figure2g.csv")
dat.birds = dat.birds |>
  mutate(difference = closer_vals - further_vals)
```

2.3 Part c

Summarize the data.

- Summarize the further data. Do the data suggest that dopamine in the brains of young zebra finches decreases when they sing further away?
- Summarize the closer data. Do the data suggest that dopamine in the brains of young zebra finches increases when they sing closer to their adult song?
- Summarize the paired differences. Do the data suggest that there is a difference between dopamine in the brains of young zebra finches when they sing further away compared to closer to their adult song?
- **Optional Challenge** Can you reproduce Figure 2(g)? Note that the you can use `geom_errorbar()` to plot the range created by adding the mean $\pm$ one standard deviation.

Solution

Our data summary suggests that dopamine in the brains of young zebra finches decreases when they sing further away ($M = -0.21$) and increases when they sing closer to their adult song ($M = 0.18$). There seems to be a difference between dopamine in the brains of young zebra finches when they sing further away compared to closer to their adult song ($M = 0.18$).

Notably, the disribution for the closer data appears to be right-skewed, suggesting that there are more higher values than lower values.

- Numerical summary

```r
closer_summary = dat.birds |>
  summarize(
    Mean = mean(closer_vals, na.rm = T),
    SD = sd(closer_vals, na.rm = T),
    Median = median(closer_vals, na.rm = T),
    IQR = IQR(closer_vals, na.rm = T),
    Skewness = e1071::skewness(closer_vals, na.rm = T),
    "Excess Kurtosis" = e1071::kurtosis(closer_vals, na.rm = T)
)

further_summary = dat.birds |>
  summarize(
    Mean = mean(further_vals, na.rm = T),
    SD = sd(further_vals, na.rm = T),
    Median = median(further_vals, na.rm = T),
    IQR = IQR(further_vals, na.rm = T),
    Skewness = e1071::skewness(further_vals, na.rm = T),
    "Excess Kurtosis" = e1071::kurtosis(further_vals, na.rm = T)
)

diff_summary = dat.birds |>
  summarize(
    Mean = mean(difference, na.rm = T),
    SD = sd(difference, na.rm = T),
    Median = median(difference, na.rm = T),
    IQR = IQR(difference, na.rm = T),
    Skewness = e1071::skewness(difference, na.rm = T),
    "Excess Kurtosis" = e1071::kurtosis(difference, na.rm = T)
)
```

```r
summary = tibble(
  Data = c("Closer Renditions", "Further Renditions", "Paired Difference"),
  Mean = c(closer_summary$Mean, further_summary$Mean, diff_summary$Mean),
  SD = c(closer_summary$SD, further_summary$SD, diff_summary$SD),
  Median = c(closer_summary$Median, further_summary$Median, diff_summary$Median),
  IQR = c(closer_summary$IQR, further_summary$IQR, diff_summary$IQR),
  Skewness = c(closer_summary$Skewness, further_summary$Skewness, diff_summary$Skewness),
  Excess_Kurtosis = c(closer_summary$Excess_Kurtosis, further_summary$Excess_Kurtosis, diff_summary$Excess_
)
```

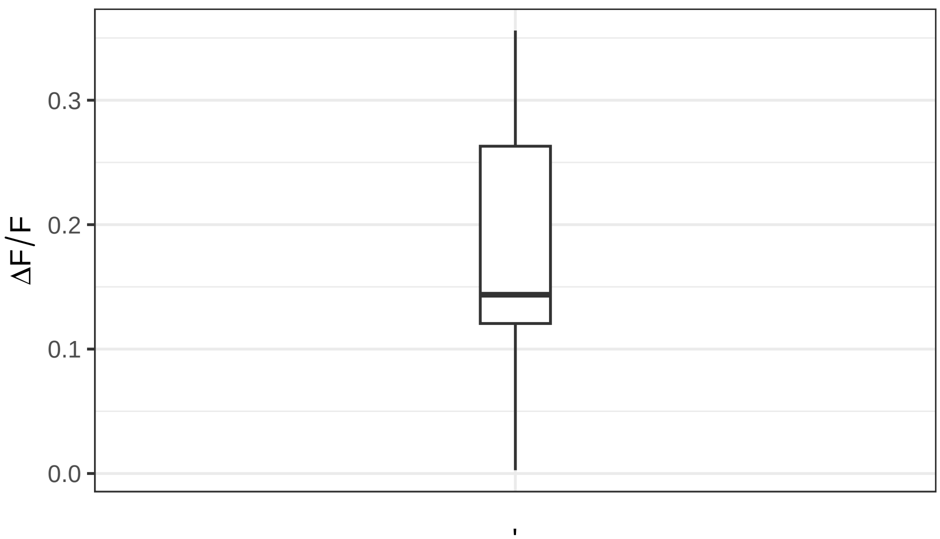
Table 1: Summary for average % change in fluorescence

Data	Mean	SD	Median	IQR	Skewness
Closer Renditions	0.18	0.10	0.14	0.14	0.26
Further Renditions	-0.21	0.13	-0.19	0.19	-1.03
Paired Difference	0.39	0.21	0.35	0.31	0.67

- Data visualization

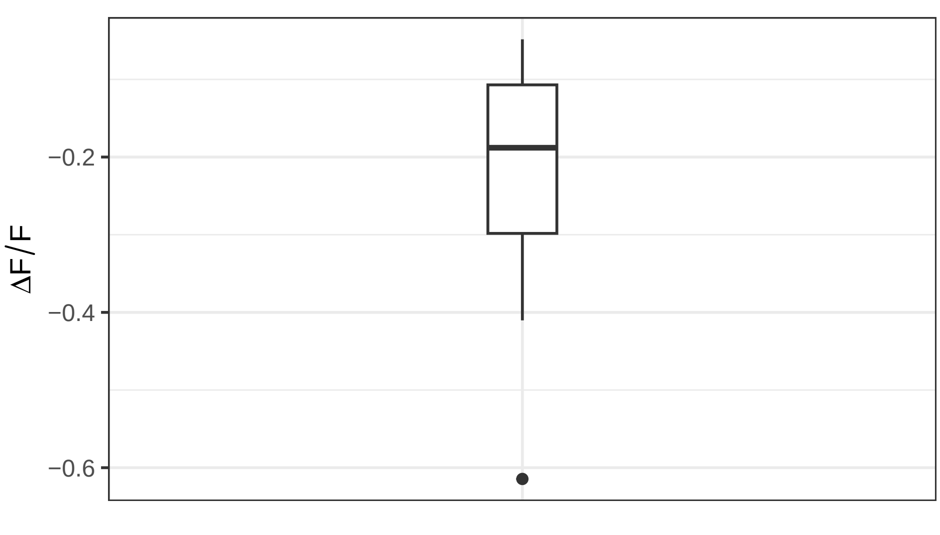
```r
ggplot(dat.birds) +
  geom_boxplot (aes (x = "",
                     y = closer_vals),
                width = 0.1)+
  theme_bw() +
  labs(x="'",
      y = expression(Delta * F/F)) +
  theme(axis.ticks.x = element_blank())
```

Figure 1: Average % change in fluorescence for closer renditions



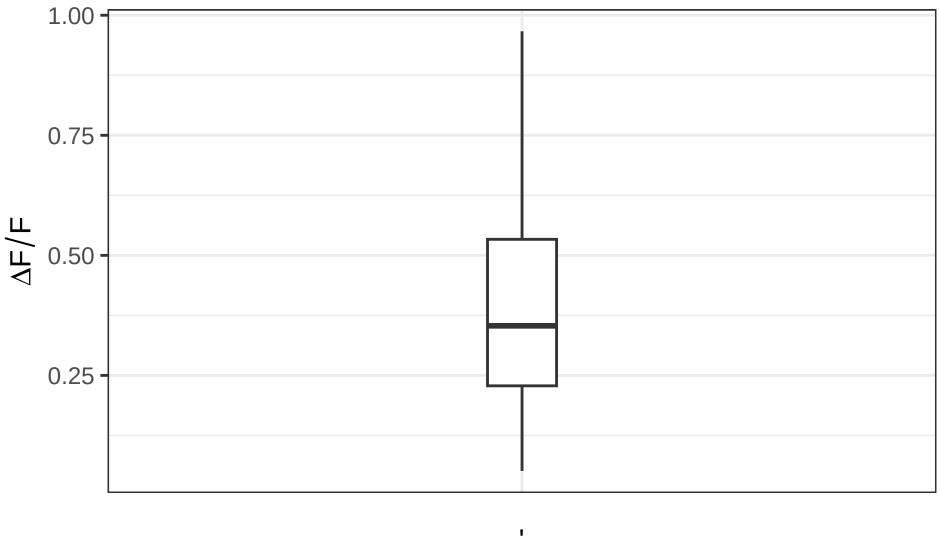

```r
ggplot(dat.birds) +
  geom_boxplot (aes (x = "",
                     y = further_vals),
                width = 0.1)+
  theme_bw() +
  labs(x="'",
      y = expression(Delta * F/F)) +
  theme(axis.ticks.x = element_blank())
```

Figure 2: Average % change in fluorescence for further renditions



```r
ggplot(dat.birds) +
  geom_boxplot (aes (x = "",
                    y = difference),
                width = 0.1)+
  theme_bw() +
  labs(x="'",
      y = expression(Delta * F/F)) +
  theme(axis.ticks.x = element_blank())
```

Figure 3: Difference in average % change in fluorescence between closer and further renditions

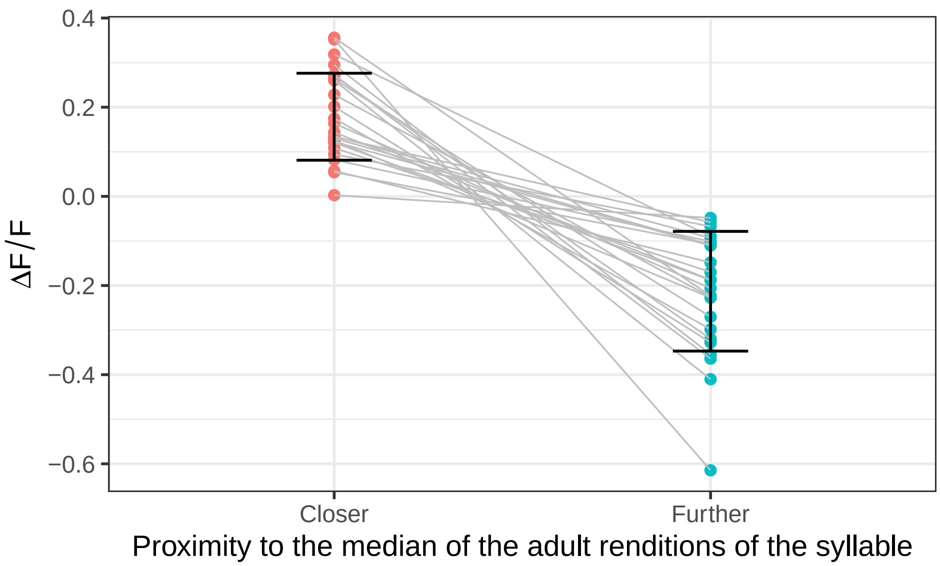


- Recreating Figure 2g:

```r
# modify dataset to plot x-axis
dat.birds.plot = dat.birds |>
  mutate(ID = seq(1, nrow(dat.birds), 1)) |>
  pivot_longer(cols = c(closer_vals, further_vals),
               names_to = "distance",
               values_to = "delta.f")

error.bars = tibble(
  distance = c("closer_vals", "further_vals"),
  upper = c(closer_summary$Mean + closer_summary$SD, further_summary$Mean + further_summary$SD),
  lower = c(closer_summary$Mean - closer_summary$SD, further_summary$Mean - further_summary$SD)
)
```

```r
ggplot()+
  geom_point(data = dat.birds.plot,
            aes(x = distance,
                y = delta.f,
                color = distance)) +
  geom_line(data = dat.birds.plot,
            aes(x = distance,
                y = delta.f,
                group = ID),
            linewidth = 0.3, color = "gray") +
# use ID to match each dot
  geom_errorbar(data = error.bars,
                aes(x = distance,
                ymin = lower, ymax = upper), width = 0.2) +
  theme_bw() +
# rename
  scale_x_discrete(labels = c("closer_vals" = "Closer",
                              "further_vals" = "Further")) +
  labs(x = "Proximity to the median of the adult renditions of the syllable",
      y= expression(Delta * F/F)) +
  guides(color = "none")
```

Figure 4: Average % change in fluorescence for closer and further renditions



2.4 Part d

Conduct the inferences they do in the paper. Make sure to report the results a little more comprehensively – that is your parenthetical should look something like: ($t = 23.99$, $p < 0.0001$; $g = 1.34$; 95% CI: 4.43, 4.60). Ensure to verify any conditions for use.
Note: Your numbers will vary slightly because they reported two-sided test p-values instead of the specified one-sided tests.

- "The close responses differed significantly from 0 ($p = 1.63 \times 10^{-8}$).''
  Note: Use the appropriate one-sided test.
- "The far responses differed significantly from 0 ($p = 5.17 \times 10^{-8}$).''
  Note: Use the appropriate one-sided test.
- "The difference between populations was significant ($p = 1.04 \times 10^{-8}$).''
  Note: Use the two-sided test.

Solution

- The close responses differed significantly from 0 ($t = 9.16$, $p < 0.0001$; $g = 1.77$; 95% CI: 0.14, 0.22).

```r
t1 = t.test(x=dat.birds$closer_vals,
      mu=0, alternative="greater")
# can't obtain CI from 1-sided test

ci1 = t.test(x=dat.birds$closer_vals,
      mu=0, alternative="two.sided")$conf.int

library(effectsize)
g1 = hedges_g(x=dat.birds$closer_vals, mu = 0)
(t1)
```

```
    One Sample t-test

data:  dat.birds$closer_vals
t = 9.1573, df = 24, p-value = 1.333e-09
alternative hypothesis: true mean is greater than 0
95 percent confidence interval:
 0.1452511       Inf
```

```
sample estimates:
mean of x
0.1786241
```

- The far responses differed significantly from 0 ($t = -7.92$, $p < 0.0001$; $g = -1.53$; 95% CI: -0.27, -0.16).

```
t2 = t.test(x=dat.birds$further_vals,
      mu=0, alternative="less")

ci2 = t.test(x=dat.birds$further_vals,
      mu=0, alternative="two.sided")$conf.int
g2 = hedges_g(x=dat.birds$further_vals, mu = 0)
(t2)
```

```

    One Sample t-test

data:  dat.birds$further_vals
t = -7.9245, df = 24, p-value = 1.866e-08
alternative hypothesis: true mean is less than 0
95 percent confidence interval:
       -Inf -0.1668619
sample estimates:
 mean of x
-0.2128063
```

- The difference between populations was statistically significant ($t = 9.14$, $p < 0.0001$; $g = 1.77$; 95% CI: 0.30, 0.48).

```
t3 = t.test(x=dat.birds$difference,
      mu=0, alternative="two.sided")
g3 = hedges_g(x=dat.birds$difference, mu = 0)
(t3)
```

```

    One Sample t-test

data:  dat.birds$difference
t = 9.1428, df = 24, p-value = 2.746e-09
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 0.3030683 0.4797923
sample estimates:
mean of x
0.3914303
```

```
stats = tibble(
  Data = c("Closer Renditions", "Further Renditions", "Paired Difference"),
  t = c(t1$statistic, t2$statistic, t3$statistic),
  p = c(t1$p.value, t2$p.value, t3$p.value),
  g = c(g1$Hedges_g, g2$Hedges_g, g3$Hedges_g),
  CI_LB = c(ci1[1], ci2[1], t3$conf.int[1]),
  CI_UB = c(ci1[2], ci2[2], t3$conf.int[2])
)
```

Table 2: Summary for average % change in fluorescence

Data	t	p	g	CI_LB	CI_UB
Closer Renditions	9.16	0	1.77	0.14	0.22
Further Renditions	-7.92	0	-1.53	-0.27	-0.16

Table 2: Summary for average % change in fluorescence

Data	t	p	g	CI_LB	CI_UB
Paired Difference	9.14	0	1.77	0.30	0.48

2.5 Part e

For each hypothesis test in the previous question, plot the null T distribution and the observed t statistic. Ensure to use `geom_ribbon()` to highlight the rejection region and the p-value. Additionally, add the resampling distribution for T to the chart to demonstrate any differences between the null distribution and the distribution suggested by the data.

Solution

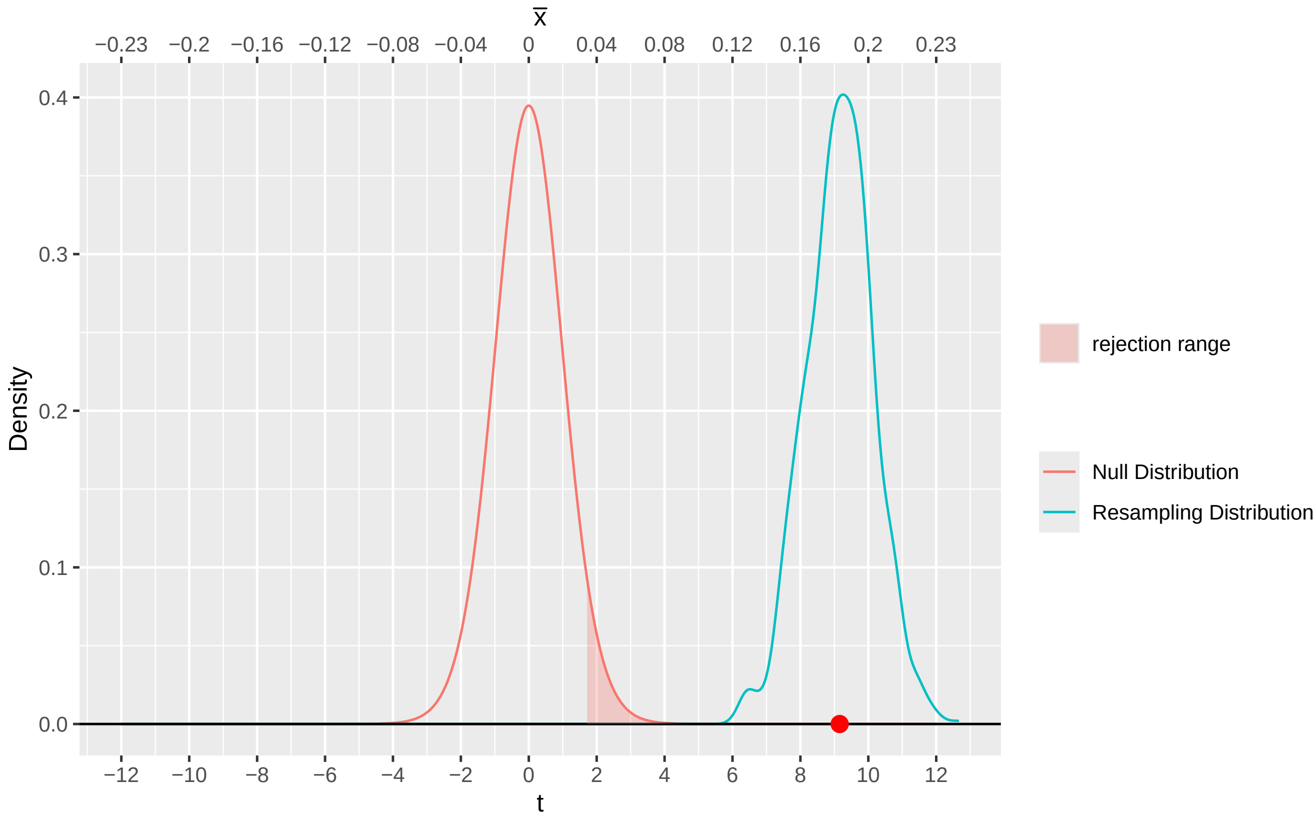
- Close responses is larger than 0.

```
mu0 <- 0
x.bar <- mean(dat.birds$closer_vals)
s <- sd(dat.birds$closer_vals)
n = nrow(dat.birds)
t.stat <- (x.bar - mu0)/(s/sqrt(n))

# critical values
alpha <- 0.05
t.critical.right <- qt(1-alpha, df=n-1)
t.critical.left <- qt(alpha, df=n-1)
t.critical.lower <- qt(alpha/2, df=n-1)
t.critical.upper <- qt(1-alpha/2, df=n-1)

# pdf.h0
ggdat <- tibble(t = seq(-12, 12, length.out=1000)) |>
    mutate(pdf.h0 = dt(t, df=n-1))

# create vectors of x-axis ticks
t.breaks <- seq(-12, 12, 2)
xbar.breaks <- round(t.breaks * (s/sqrt(n)) + mu0, 2)

# resampling for h0 distribution
ts <- rep (NA, 1000)
for (i in 1:1000){
  curr.sample <- sample(dat.birds$closer_vals, size=n, replace = T)
  ts[i] <- (mean(curr.sample) - mu0)/ (s/sqrt (n))
}
```

- The far responses is smaller than 0.

```
mu0 <- 0
x.bar <- mean(dat.birds$further_vals)
s <- sd(dat.birds$further_vals)
n = nrow(dat.birds)
t.stat <- (x.bar - mu0)/(s/sqrt(n))

# resampling for h0 distribution
ts <- rep (NA, 1000)
for (i in 1:1000){
  curr.sample <- sample(dat.birds$further_vals, size=n, replace = T)
  ts[i] <- (mean(curr.sample) - mu0)/ (s/sqrt (n))
}
```

Figure 5: Distribution of t-statistic under H_0 and H_a




```r
# resampling for h0 distribution
ts <- rep (NA, 1000)
for (i in 1:1000){
  curr.sample <- sample(dat.birds$further_vals, size=n, replace = T)
  ts[i] <- (mean(curr.sample) - mu0)/ (s/sqrt (n))
}
```

- Paired difference is significant.

```r
mu0 <- 0
x.bar <- mean(dat.birds$difference)
s <- sd(dat.birds$difference)
t.stat <- (x.bar - mu0)/(s/sqrt(n))

# resampling for h0 distribution
ts <- rep (NA, 1000)
for (i in 1:1000){
  curr.sample <- sample(dat.birds$difference, size=n, replace = T)
  ts[i] <- (mean(curr.sample) - mu0)/ (s/sqrt (n))
}
# resampling for h0 distribution
ts <- rep (NA, 1000)
for (i in 1:1000){
  curr.sample <- sample(dat.birds$difference, size=n, replace = T)
  ts[i] <- (mean(curr.sample) - mu0)/ (s/sqrt (n))
}
```

Figure 6: Distribution of t-statistic under H_0 and H_a

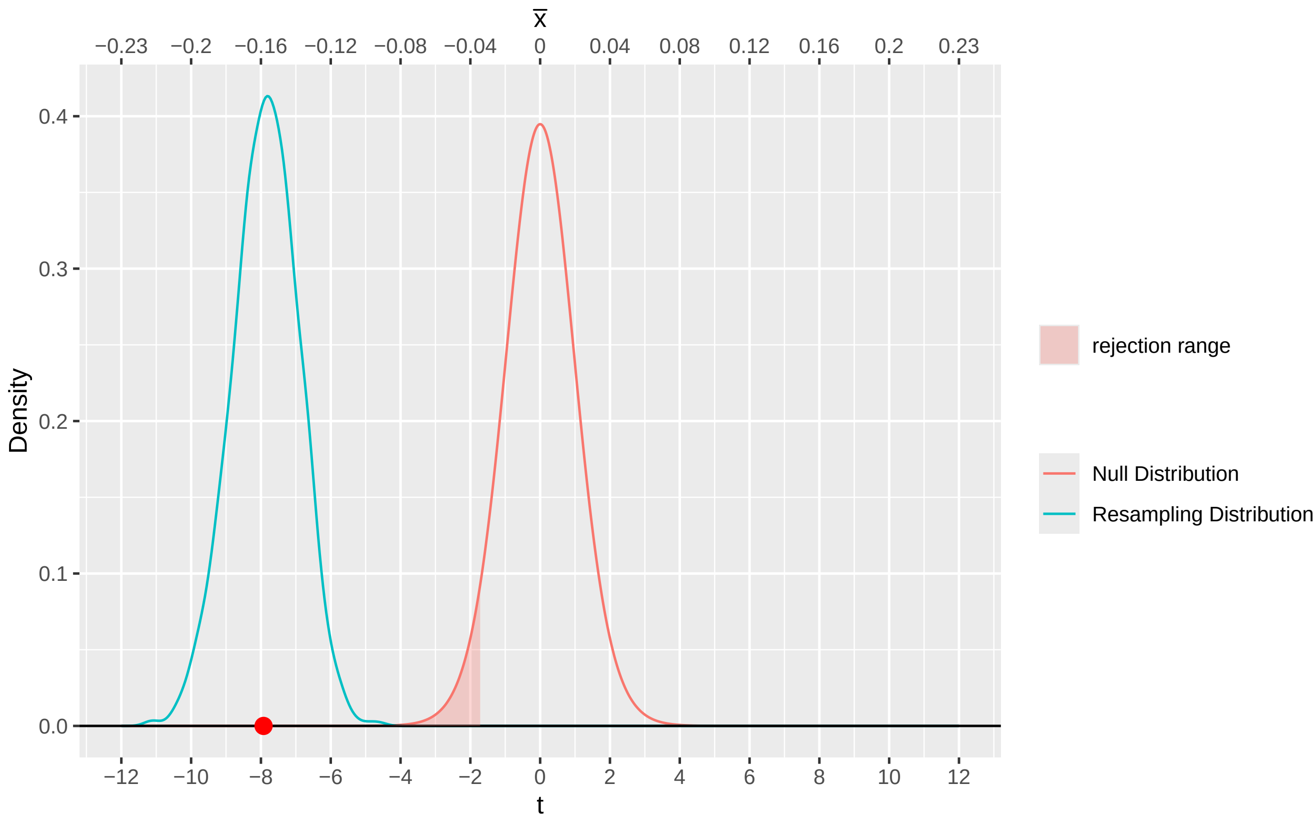
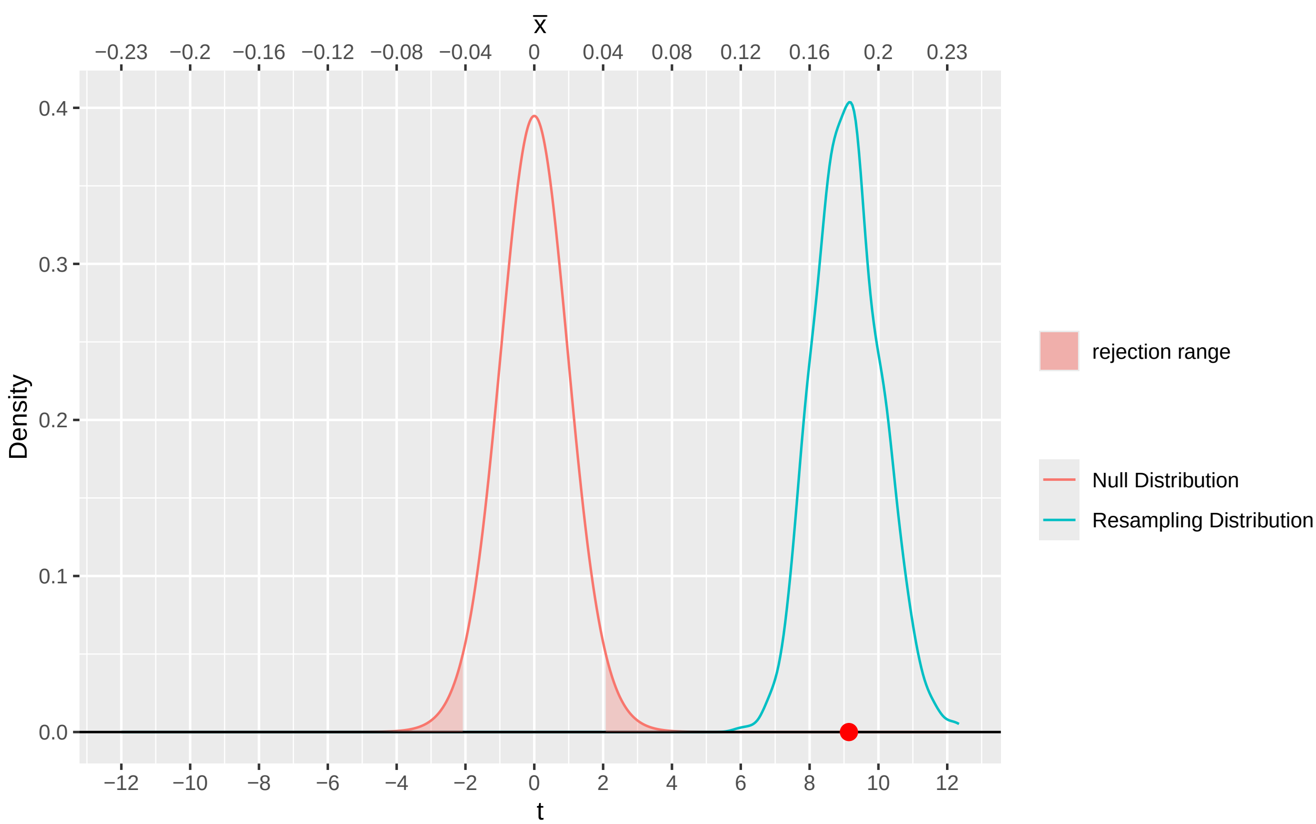


Figure 7: Distribution of t-statistic under H_0 and H_a



References

Champely, Stephane. 2020. *Pwr: Basic Functions for Power Analysis*. https://CRAN.R-project.org/package=pwr.

Kasdin, Jonathan, Alison Duffy, Nathan Nadler, et al. 2025. "Natural Behaviour Is Learned Through Dopamine-Mediated Reinforcement." *Nature*, 1–8.